

# Research Review

## sarc-authority-derivation

*Adversarial scientific and systems assessment*

**Repository:** [github.com/besanson/sarc-authority-derivation](https://github.com/besanson/sarc-authority-derivation)

Review date: 6 September 2026

**Overall assessment:** 6.7 / 10

**Central finding:** the current singleton-witness  $P^*$  is closer to a set of core attributes than a generally minimal sufficient decision reduct. Fixing this could materially strengthen the research programme.

A methodological note before the verdict: I inspected the repository structure, source, formal checkers, preregistration, generated outputs, GitHub history, releases/tags, related repositories, papers, and prior art. I was not able to perform my own clean local clone in this execution environment. GitHub-hosted CI was therefore the closest independent clean-environment execution signal available to me; importantly, current HEAD was **not green**.

There is one finding that materially changes the assessment.

## 1. Executive verdict

This is **not another agent framework**. It is a research artifact attempting something more interesting: deriving what a runtime authority gate must observe from a declared loss model and the states an agentic workflow can actually reach, rather than manually deciding the authorization schema.

That direction matters.

The repository is unusually disciplined for a project created only on **2 September 2026**: preregistration tags, frozen artifacts, machine-readable hypotheses, formal/checker separation, witnesses, deterministic outputs, proof-status discipline, explicit negative findings, and documented correction of an earlier reachability claim. There is not yet external adoption evidence.

However, there is a **central mathematical problem**.

The current definition says property  $p$  belongs to  $P^*$  when two reachable states exist that differ **only in**  $\texttt{p}$  and have different loss-model verdicts.

The paper then interprets  $P^*$  as the minimal sufficient observation set.

That implication does **not hold on constrained reachable state spaces**.

The method is effectively identifying individually indispensable — or **core** — attributes. Classical decision-reduct / discernibility theory distinguishes that from a jointly sufficient minimal reduct.

The v2 checker does not test the missing global sufficiency property; its `sound_and_minimal` flag checks recomputation identity and positive singleton witnesses.

This does not invalidate the concrete v2 output: in this particular model the nine selected properties appear sufficient because the sole excluded property, `workflow`, is genuinely irrelevant to the loss predicates. But it invalidates the **generality of the central derivation theorem**.

Ironically, fixing this could make the work substantially better.

The high-value research direction is a **governance compiler**:

*loss model + reachable execution semantics  $\rightarrow$  minimal sufficient runtime observation contract + counterexamples + enforcement schema*

That is considerably more ambitious and potentially more important than the current  $P^*$  formulation.

**Overall: 6.7/10 today.**

**Continued investment: strongly justified, but the mathematical core should be corrected before polishing or promoting the current paper.**

## 2. What the author thinks the contribution is

The intended contribution appears to be:

*Given a declared loss model  $M$ , an executable-reachable action/state space, and a set of candidate properties, mechanically derive the unique minimal set  $P^*$  that a pre-action authority gate must observe in order to reproduce  $M$ 's verdict.*

The repository then adds three supporting ideas:

- produce explicit witness pairs proving each included property matters;
- produce a `coverage_list` of candidate properties determined not to matter;
- bind authorization to a sealed, single-use execution grant.

For v2, the derivation produces nine participating properties:

- `actor_role`
- `resource_class`
- `order_value`
- `day`
- `grant_id`
- `consumed_grant_ids`
- `budget_remaining`
- `frozen`
- `min_order_quantity`

`workflow` alone is excluded.

The author additionally argues that reachability semantics can change the derived authority schema: removing the downroute transition reduces the reachable set and causes `min_order_quantity` to disappear from  $P^*$ .

That second observation is directionally important.

### 3. What I think the contribution actually is

There are really **three different contributions**, and they should be separated.

#### A. A useful governance-requirements workflow

The repository operationalizes:

**declared losses → executable predicates → reachable state model → observable-property analysis  
→ runtime gate schema**

That is useful and relatively unusual in current agentic-system engineering.

**Classification: B — novel operationalization.**

#### B. An executable finite-state diagnostic for individually indispensable authority attributes

The current participation criterion is:

$p \in P^*$  iff there exist  $x, y$  in  $R$  such that  $x$  and  $y$  differ only in  $p$  and  $M(x) \neq M(y)$

This finds properties for which there is a **singleton discernibility witness**.

That is useful, but it is not generally equivalent to finding a minimal sufficient observation set.

In rough-set terminology, the construction is very close to identifying **core attributes**, not calculating a decision reduct.

**Classification as currently claimed: E/B.**

### C. A reproducibility methodology

Preregistration before results, machine-generated paper numerals, frozen outputs, hashes, explicit claim/checker mapping, correction logs, and PROOF-STATUS labeling are excellent research-software practices.

**Classification: B/C — strong methodological and engineering contribution.**

## 4. The strongest idea

The strongest idea is **not the current participation test**.

It is:

*Authority should be synthesized from reachable consequences, rather than hand-written from an intuitive list of identity and permission attributes.*

That is potentially much bigger.

The profound systems question is not:

*“Which fields appear in my authorization policy?”*

It is:

*“What information must an enforcement point possess to distinguish every modeled executable-safe state from every modeled executable-unsafe state?”*

That turns authorization architecture into a synthesis problem.

And remediation makes the answer dynamic: a property can become authority-relevant because some transition makes a previously unreachable unsafe state reachable.

That is what the larger research programme should build around.

## 5. Author research

The evidence supports **Gaston Besanson** as the principal intellectual author.

More importantly, there is a coherent preceding body of work:

- **SARC (May 2026):** governance obligations for agents should move from prompts and post-hoc documentation into runtime architecture, with enforcement at pre-action, action-time, post-action, and escalation sites.
- **Green SARC (June 2026):** application of the architecture to resource authority — cost and carbon — with hard runtime gates contrasted against soft optimization.
- **SARC-DQ (July 2026):** application to evidence integrity and data-quality mediated action gating.
- **One Gate Is Not Enough (August 2026):** shows that remediation by one gate can invalidate judgments already made by another; therefore remediation requires re-gating and ordering semantics matter.
- **From Data Moats to Context Moats:** argues that inference-time advantage increasingly depends on governed context including memory, permissions, tools, and workflow state.
- **The Data Hydration Gap / The Inference Bottleneck:** shows a recurring preference for converting broad enterprise / AI problems into formalizable mechanisms and explicit failure conditions.

This is not a sequence of disconnected artifacts.

## 6. Intellectual trajectory

### Reconstructed trajectory

#### Enterprise governance problem



Governance cannot remain documentation



**SARC:** move constraints into the execution loop



Individual constraints have different semantics



**Green SARC / SARC-DQ:** specialize enforcement planes



Independent controls interact



**One Gate:** remediation creates stale judgments and order dependence



If gates must judge the post-remediation action...



**What must each gate actually observe?**



**Authority Derivation:** derive the observation schema from losses + reachable execution



**Next logical step:** synthesize the control plane itself from desired invariants and transition semantics.

**Is a larger programme emerging?**

**Yes.**

A strong label for it would be:

### Governance Compilation for Autonomous Systems

Core thesis:

*Agent governance should progressively move from manually authored controls toward machine-derived execution contracts compiled from explicit losses, invariants, transition semantics, authority relationships, and evidence requirements.*

This starts to look analogous to compiler construction or controller synthesis:

**governance intent → intermediate representation → enforcement architecture → executable monitors → proof obligations → traces**

That is a serious and differentiated research programme.

## 7. Novelty assessment

| Contribution                                    | Classification | Assessment                                                             |
|-------------------------------------------------|----------------|------------------------------------------------------------------------|
| Runtime governance from declared losses         | B              | Meaningful operationalization                                          |
| Reachability-conditioned authority requirements | B              | Promising agent-governance application of established formal ideas     |
| Current singleton-witness P*                    | E/B            | Useful core-attribute detector, but “minimal sufficient set” overclaim |
| Witness certificates                            | C/B            | Excellent inspectability                                               |
| Coverage list                                   | B              | Valuable governance honesty                                            |
| Downroute changes min_order_quantity relevance  | B              | Good concrete systems result                                           |
| Single-use sealed grant                         | D/C            | Established security concept, clean implementation                     |
| Preregistered machine-check pipeline            | B/C            | Strong research engineering                                            |
| Current empirical CH-A1 / CH-A3                 | D as evidence  | Primarily mechanism checks, not independent validation                 |
| Potential corrected authority-reduct compiler   | A/B potential  | Could become a real contribution if developed properly                 |

### Most important prior-art collision

The strongest missing connection is to **discernibility / decision-reduct theory**.

The current construction resembles identification of **core attributes**, whereas the true target should be a **jointly sufficient minimal reduct**.

This means the derivation theorem must be corrected or narrowed.

Other neighboring prior art includes:

- Boolean essential variables / functional dependence
- Program slicing
- ABAC policy mining
- Runtime enforcement / security automata
- Shield synthesis
- Capability systems such as Macaroons, UCAN, Biscuit

## 8. Architecture review

| Dimension           | Score | Assessment                                                                                                   |
|---------------------|-------|--------------------------------------------------------------------------------------------------------------|
| Agent architecture  | 5/10  | Clean enforcement-boundary concept; little actual agent architecture exercised here                          |
| Context engineering | 7/10  | Authority observation is treated as engineered state, but candidate/background boundaries remain handcrafted |
| Memory              | 4/10  | Stateful budget/freeze/grant history exist; durable agent memory is not studied                              |

| Dimension          | Score | Assessment                                                                         |
|--------------------|-------|------------------------------------------------------------------------------------|
| Planning           | 3/10  | Imported decisions, not planning architecture                                      |
| Tool use           | 4/10  | Relevant in spirit, lightly demonstrated                                           |
| Multi-agent design | 2/10  | Essentially outside this repository's evaluated scope                              |
| Runtime governance | 8/10  | Strong architectural placement and fail-before-effect mindset                      |
| Observability      | 8/10  | Witnesses, hashes, manifests, proof status, committed results are excellent        |
| Reliability        | 5/10  | Determinism is strong; theorem gap and failing CI reduce score                     |
| Scalability        | 4/10  | Exhaustive finite enumeration will not survive realistic high-dimensional settings |

The standout is **runtime governance**.

The weak point is scaling from finite enumeration to realistic authority schemas with many attributes, hierarchical resources, temporal state, delegation chains, and cross-field constraints.

## 9. Scientific-method review

### What is actually established

#### Within the committed finite model:

- the v2 executable model contains a finite reachable set;
- nine properties have valid singleton witness pairs;
- workflow does not;
- fresh checker recomputation agrees with the committed output;
- two differently implemented searches recover the same singleton-participation set on that same representation;
- downroute reachability makes `min_order_quantity` relevant, and removing it removes that property from the derived set.

#### Plausible

- Reachability should influence authority requirements.
- Authority schemas can profitably be derived from safety / loss models.
- Machine-produced witness certificates can make authorization requirements explainable.
- The SARC sequence is converging on a useful governance-synthesis abstraction.

#### Unsupported or currently unverified

1.  $P^*$  is generally the minimal sufficient observation set.
2. CH-A1 independently validates the derived architecture.
3. CH-A3 independently validates grant binding.
4. Experimental policy arms represent independent counterfactual executions.
5. The result generalizes to real authorization schemas.

#### Missing baselines

The strongest missing baseline is:

## decision reduct / discernibility-matrix derivation

Then compare against:

- current singleton  $P^*$
- essential-variable analysis
- ABAC policy mining
- static / dynamic dependency analysis
- symbolic safety / controller synthesis
- manually authored least-privilege policy
- solver-derived minimum observation contract

## 10. Engineering review

The research-engineering discipline is above average, but several systemic issues matter.

### Current CI is broken

HEAD and the immediately preceding v0.2 commit have failing GitHub Actions runs.

This is not evidence that the derivation itself is incorrect, but it means:

*clean-clone reproducibility is currently not demonstrated*

### CI does not execute the strongest release gate

The workflow runs bootstrap, tests and formal checks, but the broader release-check adds additional gates not obviously enforced end-to-end in CI.

### Mutation testing is not clearly a hard gate

The Makefile uses `mutmut run || true`, so the shell target does not itself clearly enforce failure on weak mutation performance.

### The repository is not packaged as a reusable library

The code is primarily “run from repository root,” not a versioned API / SDK.

### GrantLedger is a model, not production authorization

There is no distributed atomic consume, persistent transactionality, signature, issuer authentication, expiry, audience binding, or concurrency protection.

## 11. Ten ways to break the system

6. **Correlated reachable properties defeat  $P^*$ .**
7. **Missing candidate property.**
8. **Incomplete loss model.**
9. **Grid misses thresholds or interactions.**
10. **Reachability model is wrong.**
11. **Concurrent grant consumption.**
12. **Forged or stolen grant.**
13. **Combinatorial explosion.**
14. **Experimental-arm contamination.**
15. **Dependency / toolchain drift.**

The most important one is #1 because it is not hypothetical; it follows from the mathematics of the current criterion.



## 12. Questions a frontier-lab researcher would ask

16. Why is  $P^*$  a sufficient statistic for the loss verdict rather than merely the set of individually indispensable attributes?
17. What assumptions on the reachable-state relation make singleton participation equivalent to global projection sufficiency?
18. Why does the novelty analysis not engage decision reducts, discernibility matrices and rough-set core attributes?
19. Can the author characterize when the current participation algorithm and a minimum decision reduct coincide?
20. If multiple minimum observation sets exist, what does “the minimal set” mean — cardinality, inclusion, acquisition cost, privacy cost, or latency?
21. Why should all properties have equal observation cost?
22. Can reachability itself be adversarially incomplete?
23. Where is the independent oracle if the same loss predicates construct both the policy and the evaluation ground truth?
24. What changes when the loss function is probabilistic, partially observable, temporal or history-dependent?
25. How does the derivation scale beyond finite Cartesian-ish models?
26. How is the execution grant fundamentally different from capabilities such as Macaroons, UCAN or Biscuit?
27. Can semantically equivalent state representations derive different authority schemas?
28. Could authority derivation become a data-minimization guarantee?
29. Could the compiler produce enforcement placement as well as property selection?
30. What theorem would remain true if the entire procurement scenario were replaced tomorrow?

## 13. What survives the critique

Quite a lot survives.

- The architectural instinct survives.
- Reachability matters and the v2 change demonstrates that.
- Machine witnesses are highly valuable.
- The coverage-list philosophy is strong.
- The research discipline is unusually good.
- The broader research trajectory survives.

In fact, the failure of the current theorem exposes the next better research question.

## 14. Five highest-leverage interventions

| Intervention                                       | Why                                           | Impact  | Effort      | New evidence created                      |
|----------------------------------------------------|-----------------------------------------------|---------|-------------|-------------------------------------------|
| Replace $P^*$ with an Authority Reduct formulation | Fixes central theorem                         | Extreme | Medium      | Actual sufficiency/minimality certificate |
| Add a solver-backed derivation engine              | Escapes Cartesian enumeration                 | Extreme | Medium-high | Scaling curves + independent algorithm    |
| Build AuthorityBench across independent domains    | Removes procurement/loss-registry circularity | Extreme | High        | External validity                         |

| Intervention                                                     | Why                                                          | Impact | Effort     | New evidence created           |
|------------------------------------------------------------------|--------------------------------------------------------------|--------|------------|--------------------------------|
| Separate execution security from derivation theory               | Grant mechanism currently distracts and has strong prior art | High   | Low        | Cleaner novelty claim          |
| Make HEAD hermetically green and README executable in 60 seconds | Reference repos must reproduce                               | High   | Low-medium | Independent clone/run evidence |

Specifically, the current participation calculation should not be deleted — it should be renamed and situated as `core_attributes`.

The next layer should produce:

- `core_attributes`
- `all_minimal_reducts`
- `minimum_cardinality_reducts`
- `weighted_minimum_reduct`
- `redundant_attributes`
- counterexample certificates for insufficient candidate sets.

## 15. Experiments that could make the work difficult to dismiss

### Experiment A — Current-definition falsification and corrected derivation

**Hypothesis:** A discernibility/reduct-based authority compiler remains sound on arbitrary constrained reachable sets; current singleton participation does not.

### Experiment B — Representation robustness

**Hypothesis:** semantic authority requirements remain stable across semantically equivalent state representations once deterministic aliases are normalized.

### Experiment C — Cross-domain AuthorityBench

Use at least three structurally different domains:

- financial / procurement action
- source-code or infrastructure mutation
- communication / data-disclosure action

### Experiment D — Symbolic scalability

Compare:

- current enumeration
- brute-force reduct
- SAT / MaxSAT
- SMT
- optional BDD approach

### Experiment E — Authority evolution

Introduce new tools, remediators and delegated permissions incrementally and show the delta in required authority context.

## 16. Epic repository redesign

```
/
├── README.md
├── CITATION.cff
├── LICENSE
├── pyproject.toml
├── src/
│   └── sarc_authority/
│       ├── model.py
│       ├── reachability.py
│       ├── losses.py
│       ├── discernibility.py
│       ├── core.py
│       ├── reduct.py
│       ├── weighted_reduct.py
│       ├── certificates.py
│       └── grants/
├── specs/
├── examples/
├── benchmarks/
│   └── authoritybench/
├── experiments/
├── formal/
├── evals/
├── adversarial/
├── tests/
├── paper/
└── reproducibility/
```

The repository should foreground the reusable research abstraction, not the paper-production machinery.

## 17. README strategy

Recommended sequence:

31. One-line thesis
32. Problem
33. Why existing approaches are insufficient
34. Contribution
35. Architecture
36. 60-second example
37. Empirical evidence
38. Baselines
39. Reproduction
40. Limitations
41. Citation
42. Roadmap

### Strongest one-sentence thesis

***An agent should not be granted authority based on a hand-written list of attributes: given a declared loss model and reachable execution semantics, we can derive and verify the smallest runtime observation contracts sufficient to distinguish every modeled safe action from every modeled unsafe one.***

That sentence becomes defensible only after implementing the reduct formulation.

## 18. Research roadmap

### Next 30 days

- Withdraw or narrow Proposition 1.
- Add the correlated-property counterexample.
- Reframe current  $P^*$  as `core_attributes`.
- Implement exact decision reducts.
- Show whether current  $v_2$ 's core also happens to be a reduct.
- Expand literature positioning.
- Separate grant contribution from derivation novelty.
- Fix CI.
- Rewrite README.

### Next 90 days

Build **AuthorityBench v0.1** with at least three domains and add:

- SAT / SMT derivation
- minimum-cardinality and weighted reducts
- acquisition-cost-aware authority
- representation perturbations
- reachability perturbations
- external loss oracle
- separate counterfactual rollouts
- stronger baselines

### Next 6 months

Turn this from “Paper 5 of SARC” into a reusable research substrate:

## SARC Authority Compiler

Input:

- losses / invariants
- action schema
- transition / reachability semantics
- policy configuration
- observation costs

Output:

- minimum authority observation contract
- alternative equivalent contracts
- core attributes
- coverage gaps
- enforcement predicates
- witness / counterexample certificates
- change impact when transitions evolve

## 19. Publication potential

### Current state

- **Technical report:** Yes
- **arXiv paper:** Yes, after correcting Proposition 1
- **Workshop paper:** Yes
- **Conference paper:** Not yet
- **Systems paper:** Potentially
- **Benchmark paper:** Not yet
- **Open-source reference architecture:** High potential

The strongest publication path is not:

*“We discovered which nine attributes SARC authority needs.”*

It is:

*“Synthesizing Runtime Authority Observation Contracts for Autonomous Agents.”*

To get there, the work needs:

- corrected theorem
- theorem characterizing when singleton-core = reduct
- symbolic implementation
- 3+ domains
- realistic baselines
- scalability
- change-under-remediation experiment
- external reproduction

## 20. The hidden opportunity

The author appears one conceptual step away from moving from **governance-by-architecture** to **governance-by-synthesis**.

SARC manually establishes where controls belong.

Green SARC and SARC-DQ establish different classes of control.

One Gate establishes composition semantics.

This repository asks which information an authority control requires.

The logical completion is:

**Risk / Loss specification → Reachable semantics → Required observations → Control predicates → Enforcement placement → Capability scope → Runtime monitor → proof / counterexample**

In other words:

### **Compile governance intent into an executable control plane.**

A further extension would optimize not just for the fewest attributes, but the **least-cost sufficient authority context** balancing latency, privacy, acquisition cost, and staleness risk.

That connects directly back to the author’s broader “context moat” perspective.

## 21. Final verdict

**The repository matters if...** it becomes a compiler from losses and reachable execution semantics into provably sufficient, minimal runtime authority contracts rather than remaining a finite-grid demonstration of a manually chosen loss model.

**The repository fails if...** it continues calling the current singleton-witness  $P^*$  a generally minimal sufficient observation set after the core-versus-reduct distinction is known.

**The author should stop doing...** incremental paper polishing, additional terminology, and more demonstrations whose success follows structurally from the same predicates used to construct them.

**The author should double down on...** governance synthesis: correct the mathematics, derive authority reducts symbolically, expose counterexamples and alternative minimal contracts, and turn the SARC line of work into a machine-checkable compiler from governance intent to runtime enforcement.

# Appendix - Research Map

How the broader Gaston Besanson research programme connects across agentic AI, runtime governance, context, and control-plane synthesis.

